Earthquake Crack Detection From Aerial Images Using a Deformable Convolutional Neural Network

Dawen Yu, Shunping Ji, *Senior Member, IEEE*, Xue Li, Zhaode Yuan, and Chaoyong Shen

***Abstract*— Detecting the terrain surface cracks caused by earthquakes, which are termed coseismic ruptures, has important significance for discovering concealed faults, monitoring their movements, and forecasting possible follow-on earthquakes. On May 22, 2021, Maduo County in Qinghai province, China, suffered an earthquake with a magnitude of 7.4, which created densely distributed cracks. In this study, we designed an automatic crack detection framework based on remote sensing technology. With the use of low-altitude unmanned aerial vehicles (UAVs), we obtained very high-resolution aerial images of the area affected by the earthquake, which were further processed by photogrammetric software to produce digital orthophoto maps (DOMs). We then designed a novel terrain surface crack detection neural network, which differs from the previous methods that focus on detecting cracks in man-made object surfaces such as flat roads. We investigated the spatial property of the sinuous linear cracks and handled this by introducing adaptive deformable convolutions with a context-channel-space boosted mechanism. The feature extraction stage, feature optimization stage, and upsampling stage were embedded with the deformable convolutions to form a compact and powerful crack detector, named Crack-CADNet [the Context-chAnnel-space boosted Deformable convolutional neural network (CNN) for crack detection]. The postprocessing included filtering out the nontectonic cracks, aided by annotations from experts, and grouping and vectorizing the generated binary segmentation map as crack polygons, which were evaluated at the instance level. In addition to the first in-depth investigation of detecting earthquake cracks with aerial remote sensing and a deep learning-based process, the crack detection network we propose outperformed the recent CNN-based methods designed for general semantic segmentation and crack detection. Source code and the Maduo earthquake crack dataset will be available at http://gpcv.whu.edu.cn/data/.**

***Index Terms*— Deformable convolutional neural network (CNN), earthquake crack detection, remote sensing images, semantic segmentation, spatial attention.**

I. Introduction

UNDERGROUND tectonic movement can cause faults, release vast amounts of energy, and trigger an earthquake. When the earthquake has a high magnitude, coseismic ruptures commonly appear on the terrain surface, in the form of Earth surface cracks, scarps, bulges, landslides, barrier lakes, and other structures. In this study, our focus was the Earth surface cracks caused by earthquakes. On May 22, 2021, an earthquake in Maduo County, Qinghai province, China, occurred, which was the highest magnitude earthquake since the Wenchuan earthquake that caused huge damage in 2008 in China [1]. The Maduo earthquake created densely distributed cracks in the Earth's surface [see Fig. 1(d)]. Recognizing and delineating these terrain surface cracks caused by earthquakes in a timely manner is of great value for the study of fault activity and tectonics. The reason for this is that active faults can further trigger earthquakes, landslides, and other geological disasters, which can cause secondary damage to people's lives and properties. Currently, earthquake crack detection is a manual task; in this article, we propose a complete and automatic pipeline for earthquake crack detection. We describe the remote sensing data acquisition conducted in this study, the preprocessing, and the development of a new deep learning-based method for the fast and automatic detection of earthquake cracks. The postprocessing conducted

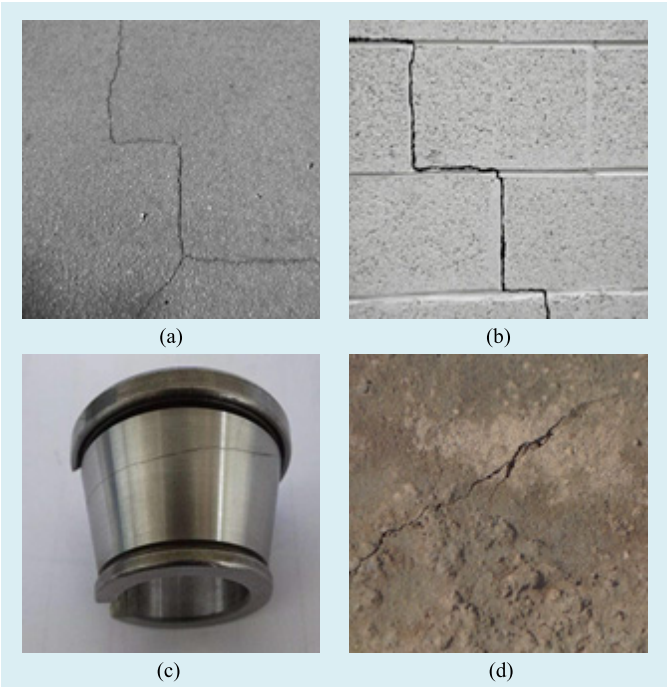


Fig. 1. Cracks in a (a) road, (b) wall, (c) industrial component, and (d) natural terrain surface.

Manuscript received April 19, 2022; revised June 4, 2022; accepted June 10, 2022. Date of publication June 15, 2022; date of current version June 24, 2022. This work was supported by the National Natural Science Foundation of China under Grant 42171430 and Grant 42030102. *(Corresponding author: Shunping Ji.)*

Dawen Yu and Shunping Ji are with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430079, China (e-mail: yudawen@whu.edu.cn; jishunping@whu.edu.cn).

Xue Li is with the Institute of Seismology, China Earthquake Administration, Wuhan 430072, China (e-mail: leexue1211@126.com).

Zhaode Yuan is with the State Key Laboratory of Earthquake Dynamics, Institute of Geology, China Earthquake Administration, Beijing 100029, China (e-mail: yzd19862922@163.com).

Chaoyong Shen is with The Third Surveying and Mapping Institute of Guizhou Province, Guiyang 550004, China (e-mail: shency@gzchsy.cn).

Digital Object Identifier 10.1109/TGRS.2022.3183157

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:04:21 UTC from IEEE Xplore. Restrictions apply.

for excluding nonearthquake cracks and the instance-level performance evaluation is also described.

Although we are not aware of any deep learning-based methods especially designed for earthquake crack detection from remote sensing imagery, the detection of other types of cracks has been studied [2]–[8]. Cracks can appear on the faces of various man-made structures, e.g., pavements [2], [3], bridges [4], buildings [5], walls [6], and other civil structures [7], [8], with linear or curvilinear characteristics. The various properties of these cracks, including the length, width, angle, topology, and density, can be measured after crack detection to assess the extent of the damage [9]–[12]. Deep learning-based methods, and especially convolutional neural network (CNN)-based methods, have been used to automatically extract the features of cracks [13], [14]. Nevertheless, differing from cracks in smooth and homogeneous structures, earthquake cracks in uneven terrain surfaces with noisy backgrounds can be harder to detect accurately, as shown in Fig. 1. In the following parts of this section, we introduce the recent crack detection methods developed for use with man-made object surfaces, which are rooted in general CNN-based methods. We then briefly introduce our work and the contributions of this study.

A. Crack Detection in Man-Made Objects

The recently developed CNN-based methods detect cracks in man-made object surfaces at either the patch level or the pixel level.

Zhang *et al.* [15] developed a deep CNN-based method to classify image patches with or without cracks in road images captured by smartphone, which was reported to work better than the conventional methods. Pauly *et al.* [16] also reported that deeper networks can achieve a superior accuracy in pavement crack detection. The patch-based classification methods generally obtain a lower accuracy than the pixel-wise segmentation methods, so the latter are now the mainstream methods, most of which use a fully convolutional network (FCN) [17] as the backbone structure.

Zou *et al.* [18] built a network called DeepCrack based on SegNet [19] for segmenting cracks in road pavements and stones from close-range images. Dung [20] proposed an FCN for extracting cracks in various campus buildings in close-range images, and compared three different encoders. Liu *et al.* [21] designed a deep network called DeepCrack, which applies multilevel feature aggregation and multiscale supervision for detecting asphalt/concrete cracks from Internet images and personally taken photos. Li *et al.* [22] proposed the skip connected crack detection network (SCCDNet) to conduct pixel-level crack segmentation in self-collected close-range images, with a skip-squeeze-and-excitation module and depth-wise separable convolution units. Similar works can be found in [23]–[26].

Although there is an increasing trend of applying CNN-based methods in man-made object surface crack detection from close-range images including detecting the infrastructure damages after earthquakes [27], these methods have not yet been employed to detect cracks in natural terrain surfaces. More importantly, these studies are restricted to methodology studies about image-based classification or pixel-level segmentation. A complete engineering-based crack detection pipeline is currently missing. However, there is an urgent requirement to be able to detect the cracks in the Earth's surface after an earthquake, which was the case after the Maduo earthquake. This situation impelled us to explore and develop a complete earthquake crack detection pipeline, including data acquisition and preprocessing, a suitable CNN-based method, and postprocessing, for quick and highly accurate crack detection.

B. Proposed Crack Detection Pipeline

In this study, we proposed an earthquake crake detection pipeline with three main parts. In the first part, we used unmanned aerial vehicles (UAVs) to capture scenes of the whole region affected by the Maduo earthquake, and then processed the individual images into large digital orthophoto maps (DOMs) to be used in the next step with photogrammetry technology. Finally, all of the cracks in the DOMs were delineated by extensive manual annotation, which were then used by seismologists for evaluating the stability of the faults in the area and as samples for the supervised deep learning method.

The second part was to develop a specific earthquake crack detection method for use with aerial imagery, which was the main work in this study. We took the orthophoto images as input, designed a suitable network, which was based on the recent development in FCN-based methods, and trained the network with the manually delineated cracks. Finally, we predicted the cracks in a test area. Please note that the cracks were sinuous and very thin [see Fig. 1(d)], so their labels were manually delineated as vector-form lines. However, the current CNN-based methods prefer raster images. We buffered the vector-form lines with a rational small width, which were then used for the training and performance evaluation.

The third part involved processing the pixel-level segmentation map from the FCN network into a crack instance map by grouping the adjacent foreground pixels. A crack instance map provides richer information for seismologists, such as the number of cracks and the distribution of the largest cracks. The other thing to be considered in the postprocessing is to exclude nontectonic cracks on the Earth's surface. In fact, there is no obvious difference in aerial remote sensing images between the coseismic ruptures caused by a recent earthquake and nontectonic cracks in the Earth's surface caused by dissolution, weathering, thermal expansion, cold shrinkage, or water loss. This causes difficulties in crack detection for both humans and algorithms. Nevertheless, geologists can locate earthquake cracks based on the information of the epicenter location, the seismic zone, and the magnitude of the destructive earthquake. We incorporated the boundary of the earthquake rupture region identified by human experts to exclude cracks that were not coseismic ruptures.

When designing the crack detection method, we noted that previous studies have used regular convolutions to extract the features of various cracks. However, the special sinuous

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:04:21 UTC from IEEE Xplore. Restrictions apply.

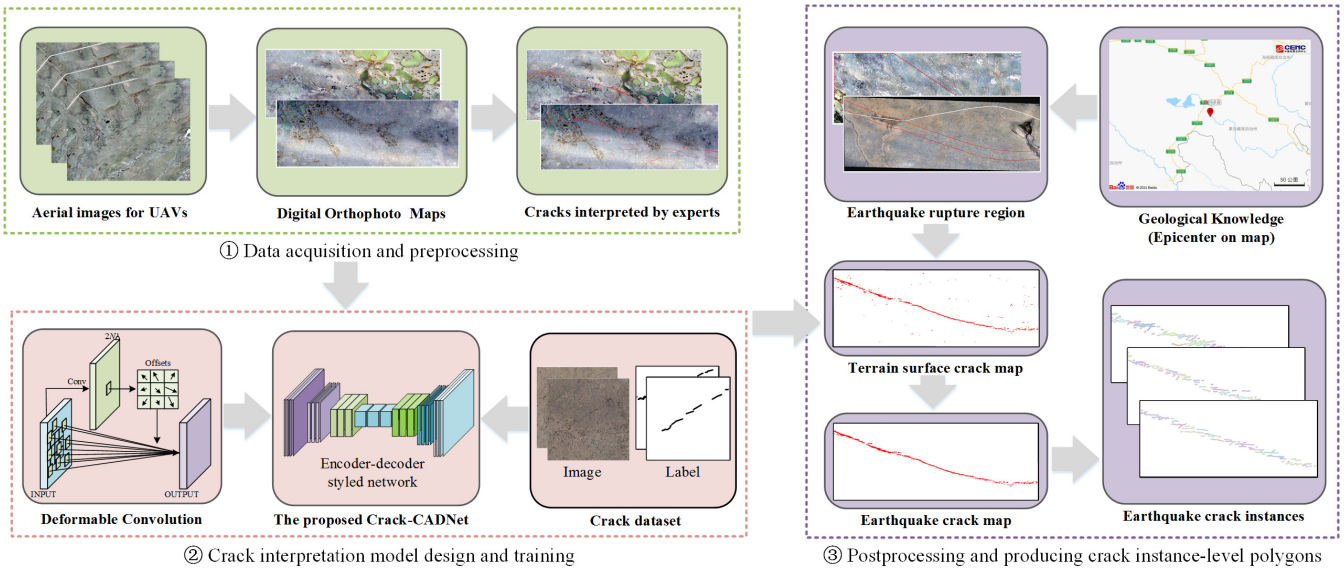


Fig. 2. Whole workflow of earthquake crack detection, from data acquisition to automatic interpretation by algorithm. In the subgraph of the earthquake crack instances, the different colored polygons indicate different individual crack instances.

and irregular shapes of cracks were not considered. In this article, we argue that these spatial characteristics of cracks should be taken into consideration. Regular grid sampling with standard convolutions is likely to be less adaptive to the sinuous pattern of cracks. Therefore, we instead introduced deformable convolutions [28] to improve the performance of crack detection. Specifically, the deformable convolutions add extra 2-D offsets into the common convolution sampling grid, i.e., the locations of the sampling are changeable, which benefits the extraction of irregularly shaped objects. The deformable convolutions were embedded in the feature extractor, the feature optimization stage, and the upsampling stage in the designed crack detection network, which we called Crack-CADNet. It should be noted that we do not employ the instance segmentation methods because many adjacent cracks are practically difficult to separate. Moreover, the strip-shaped cracks are not very suitable for instance segmentation, just like roads, instead of buildings or ships [29]. Therefore, the proposed Crack-CADNet treats the crack detection as a semantic segmentation task.

C. Contributions

The main contributions of this work are summarized as follows.

1) A complete deep learning-based earthquake crack detection pipeline is proposed for the first time, which was applied in post-disaster management of the recent Maduo earthquake.
2) The spatial shape property of terrain cracks was investigated using a deep learning framework. Deformable convolutions were applied to improve the crack detection performance. A novel neural network architecture was then designed for terrain surface earthquake crack detection.
3) For the crack detection tasks in both natural terrain surfaces and the earthquake rupture region labeled by human experts in Maduo and the adjacent region, the proposed method comprehensively and significantly outperformed the state-of-the-art crack detection methods.
4) We edited and released a high-resolution earthquake crack dataset (http://gpcv.whu.edu.cn/data/), the first open-source aerial earthquake crack dataset called the Maduo dataset.

II. Method

In this section, the pipeline of earthquake crack detection is first presented. The proposed crack detection network and all of its components are then detailed, including the feature extraction sub-network, the feature optimization sub-network, the upsampling sub-network, and the adopted loss function. Finally, the postprocessing strategy used for obtaining instance-level earthquake cracks is provided.

A. Pipeline of Earthquake Crack Detection

The pipeline consists of data acquisition and preprocessing, crack detection, and postprocessing, as shown in Fig. 2. The first part includes aerial remote sensing data capture, DOM generation, and manual crack interpretation. To access a wide range of earthquake cracks, images from low-altitude aerial photography are the most suitable data source, as satellite images can miss a large amount of the fine cracks. In this study, JOUAV CW-15 UAVs equipped with professional Sony ILCE-7RM2 cameras were employed to capture images right after the Maduo earthquake event. These images are the first very high resolution remote sensing data covering the earthquake area. A total of 51 054 images with a 70%/70% overlap rate were obtained. The images were then processed with photogrammetry technology to produce orthophoto maps.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:04:21 UTC from IEEE Xplore. Restrictions apply.

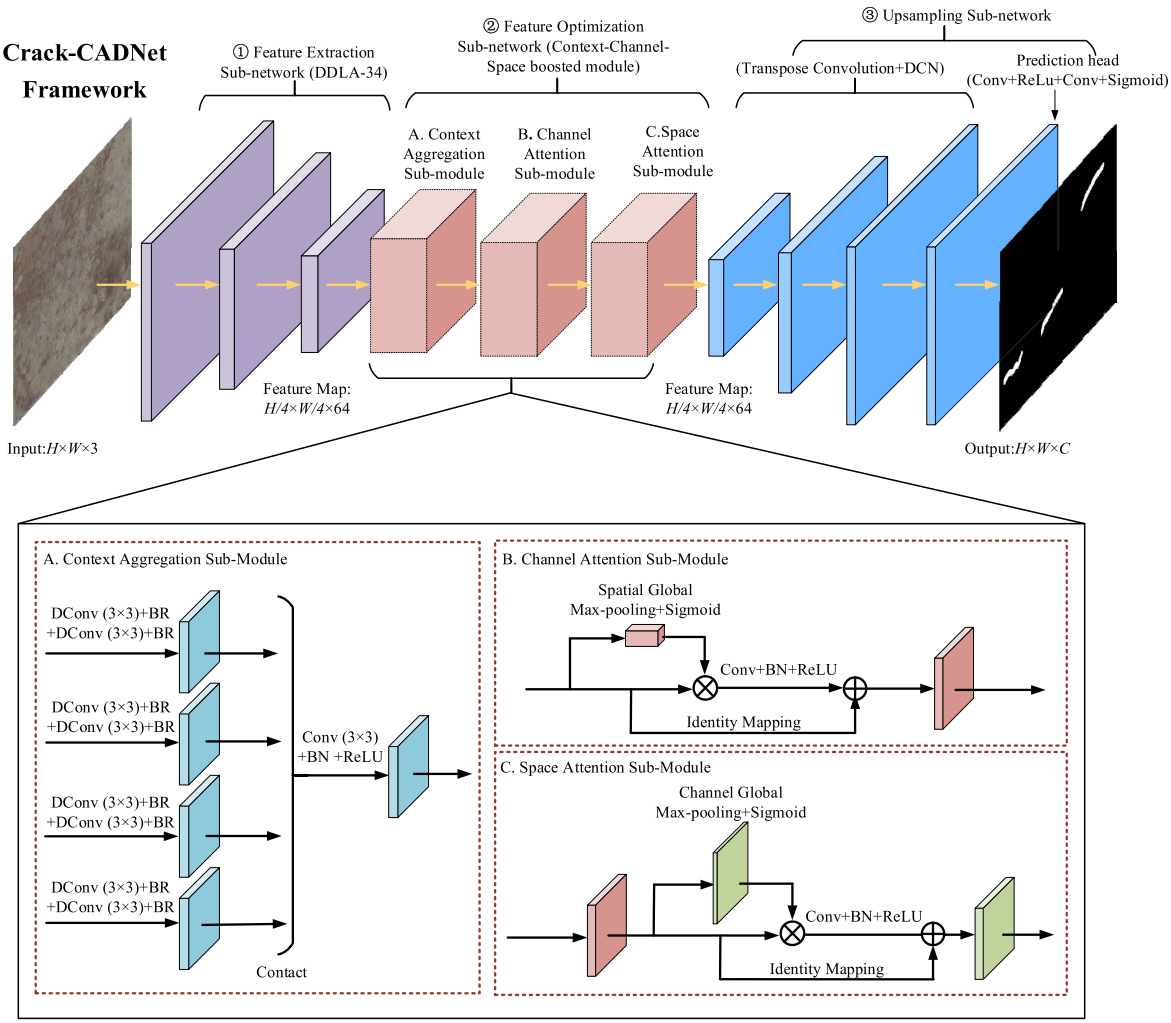


Fig. 3. Proposed context-channel-space boosted deformable convolutional neural network for natural terrain surface crack detection (Crack-CADNet).

In total, we produced 382 DOMs covering the whole region affected by the Maduo earthquake. Experts interpreted the cracks in the DOMs one by one, and a careful cross-check was carried out to avoid omission errors and incorrect annotations. It took two human experts four months to interpret all the data, and the cracks were recorded as vector-form lines. As mentioned above, we expanded the single-pixel lines to thin polygons by a buffering operation as a one-pixel line provides few cues for a CNN model to learn crack features and converge.

The second part of the pipeline is the crack detection network we developed. We designed an adaptive and effective crack interpretation model, which takes the spatial properties of sinuous cracks into consideration. In particular, we embedded deformable convolutions into all three stages of the model, which results in a significant improvement in crack delineation performance (see Section II-B).

The third part of the pipeline, i.e., the postprocessing, involves nontectonic crack elimination and instance-level crack production, which are detailed in Section II-C. Extra geological knowledge is introduced to remove the nontectonic cracks that are detected by the network, and finally, the remaining cracks are grouped and vectorized into individual crack polygons.

B. Proposed Crack Detection Network Architecture

We developed an effective framework for detecting terrain surface cracks from remote sensing images, which we called Crack-CADNet (the Context-chAnnel-space boosted Deformable CNN for crack detection). As shown in Fig. 3, Crack-CADNet contains three components, i.e., a feature extraction sub-network, a feature optimization sub-network, and an upsampling sub-network.

Firstly, the feature extraction sub-network is based on the deep layer aggregation (DLA) network [30] with deformable convolution layers and more skip connections. The deformable convolutions help the sub-network adapt to the crack-shaped spatial characteristics, and the skip connections are for the multiscale semantic information. The feature optimization sub-network is designed to adjust the intermediate feature maps by a self-attention mechanism that recalibrates the semantic information of the multishape cracks in both the spatial and channel dimensions. Finally, the upsampling sub-network

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:04:21 UTC from IEEE Xplore. Restrictions apply.

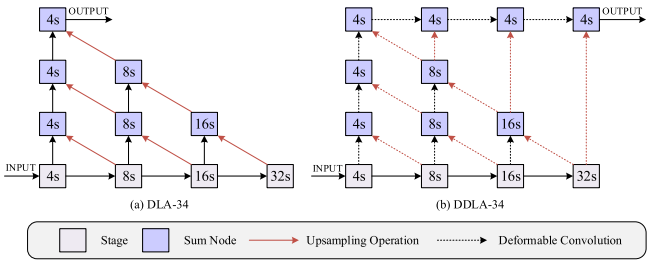


Fig. 4. Difference between (a) the original DLA-34 and (b) the version using deformable convolutions, i.e., DDLA-34. The numbers in the boxes represent the downsampling strides to the input image.

restores the downsampled features to the original scale with transpose convolutions and deformable convolutions. At the end of the upsampling sub-network, a prediction head maps the features to the final binary prediction results.

1) Deformable Convolution: A deformable convolution can be formulated as

$$y(p_0) = \sum_{p_n \in R} w(p_n) \cdot x(p_0 + p_n + \Delta p_n). \tag{1}$$

Given the input feature map x, the output convolved feature at the location of p_0 is calculated by weighting the input features at the locations of $(p_0 + p_n +_\Delta p_n)$, where $p_n \in \{(-1, -1), (-1, 0), (-1, 1), \ldots, (1, -1), (1, 0), (1, 1)\}$ in a 3×3 convolution, and Δp_n represents the learned offset. The bias term is omitted, for the sake of simplicity. Note that Δp_n is typically fractional, so bilinear interpolation is employed to obtain the new feature value at the position of $(p_0 + p_n + \Delta p_n)$.

2) Feature Extraction Sub-Network: DLA [30] is a compact and widely used feature extractor backbone in image classification and semantic segmentation. DLA (DLA-34 in our study) iteratively merges the hierarchical feature maps for better accuracy and fewer parameters. As shown in Fig. 4, we used the deformable convolutions to replace the common convolutions and added three additional unsampling skip connections. This was used as backbone in the previous object detection study [31] and we called it DDLA-34 here. We experimentally found that this encoder is also highly adaptable to crack segmentation, and can obtain better effects than DLA-34.

3) Feature Optimization Sub-Network: The attention mechanism has been widely applied in natural image processing and remote sensing object recognization [32]–[34], which emphasizes the foreground and suppresses the background by self-recalibrating the inner feature map on spatial and/or channel dimensions. After the fused features across hierarchical layers are extracted by the DDLA-34 encoder, we introduce an adaptive context-channel-space boosted module for highlighting the information of the foreground cracks and suppressing the interference of the noisy background. The module consists of concatenated context, space, and channel sub-modules. The context aggregation module handles scene context information with four 3×3 parallel and weight-unshared deformable convolutional layers, as shown in Fig. 3, providing a comprehensive description of a current pixel. The parallel deformable convolution groups expand the network width and have been proved very effective in our crack detection task. Following the context sub module, a channel attention sub module and a space attention sub module and are applied in sequence. In the space and channel modules, global max-pooling and sigmoid activation are employed, respectively, to generate the final weight map. The recalibrated features are finally obtained by multiplying the original feature map by the weight map. The weighted features are then further convolved and element-wise added with the original features. Details are provided in Fig. 3. Other structures of attention modules and parallel mode sub-modules were experimentally tried; however, we found that the concatenation of context-channel-space module performed the best.

4) Upsampling Sub-Network: The upsampling sub-network is designed to decode the four $\times$ downsampled features and to generate the final binary (crack versus noncrack) segmentation map at the original scale. In previous studies, transpose convolutions (or deconvolutions) and bilinear interpolation have been widely employed to restore the feature maps to the original image size [35]–[39]. In the proposed method, we adopt learnable transpose convolutions. In particular, we add a 3×3 deformable convolutional layer after each transpose convolutional layer to restore the irregular shapes of the cracks. Two groups of transpose and deformable convolution combinations in the upsampling stage with a stride of 2 are adopted. Finally, the feature maps are processed by a convolutional layer and a rectified linear unit (ReLU), and then a convolutional layer and a sigmoid activation function are used to produce the probability map. A threshold of 0.5 is used to discriminate cracks and background.

5) Loss Function: Binary cross-entropy (BCE) loss is adopted to measure the prediction error in the segmentation map. As the background pixels occupy the majority, we add a larger weight factor w for positive samples in the BCE loss function to alleviate the imbalance, which is defined as

$$\text{loss}(p_x) = \begin{cases} -\log(1 - p_x), & \text{if } p_y = 0 \\ -w \cdot \log(p_x), & \text{if } p_y = 1 \end{cases} \tag{2}$$

where p_x and p_y are the prediction and label, respectively. 0 represents a background pixel and 1 is a crack pixel. The total loss is defined as the average value of all the losses in the whole prediction map. After trial and error, we used $w = 10$ in our study.

C. Postprocessing

Considering the limitation of GPU capacity, the original remote sensing images had to be cropped into smaller image tiles (512×512 pixels in our case) and then fed into the CNN. As a result, some cracks were inevitably cut into parts in the different tiles. To improve the precision and obtain complete crack instances, we applied an overlapping cropping strategy with a 1/4 overlap ratio. The 1/8 border of each tile of the prediction map was abandoned to avoid the use of unreliable results in the border pixels. Finally, we stitched all the predictions of the tiles to form the large-size prediction map. This process was designed in an online way. After

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:04:21 UTC from IEEE Xplore. Restrictions apply.

obtaining the pixel-level crack segmentation maps, we transformed them into instance-level polygons by a vectorization operation.

The whole process presented above can be used to predict natural terrain cracks, but it cannot distinguish nontectonic cracks and earthquake cracks, which have similar features in the imagery. For this task, extra geological knowledge is required. In this work, we restricted the prediction results to a given region, which was interpreted by experts analyzing the seismotectonics and seismicity information of Maduo region. By filtering out the detection results that did not belong to the earthquake rupture region (many of which were false positive detections), the earthquake crack extraction performance was significantly improved.

III. Experiments and Results

The details of the study area and the dataset we built are provided in Section III-A; the evaluation metrics and experimental settings are introduced in Section III-B; the results of the different methods are provided in Section III-C; the ablation experiments are reported in Section III-D; and a discussion about the efficiency, limitation, and applications of the proposed method is finally presented in Section III-E.

A. Study Area and Dataset Preparation

The study area was Maduo County and the adjacent Maqin County, north of the Tibetan Autonomous Prefecture of Golog, Qinghai province, China, which we refer to here as Maduo for short. The geotectonic unit in this area belongs to the Bayan Kara fold belt, and the structural lines stretch from northwest to southeast. The dominant geomorphic types in Maduo are wide valleys and river/lake basins, which are mostly formed by faulting. There are also level terrain regions, deserts, and swamps. On May 22, 2021, an earthquake with a magnitude of 7.4 occurred and produced many earth surface cracks.

From May 24 to June 14, 2021, we used JOUAV CW-15 UAVs equipped with Sony ILCE-7RM2 cameras to capture the images of the study area, which were then processed using PhotoScan software to produce 382 DOMs.

We selected 12 representative DOMs, in which cracks were concentrated, to create the Maduo terrain surface crack dataset (the Maduo crack dataset for short), which was carefully edited and checked, for developing the supervised deep learning-based methods. The size of the DOMs varied from 21 781 × 14 740 pixels to 39 064 × 14 926 pixels, with a ground sampling distance of around 0.1 m. We divided the dataset into a training set and a test set by the popular partition ratio of 2:1. Four DOMs from the test set and two of the eight DOMs in the training set are shown in Fig. 5. The terrain surface cracks in the DOMs of the dataset were interpreted and annotated by experts from the China Earthquake Administration, using ArcGIS software, and all the crack instances were recorded as vector-form lines and recorded in the shapefile. There were a total of 12 065 earthquake cracks and 1174 nontectonic cracks in the experimental images. Some of the human-annotated earthquake crack lines are shown in Fig. 6.

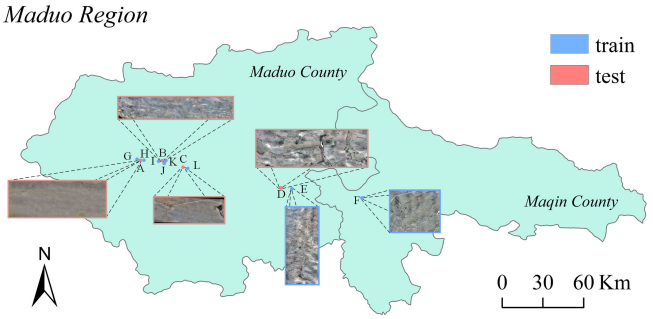


Fig. 5. Geographical distribution of the training and test images in the Maduo terrain surface crack dataset. The blue/pink points provide the real geographical locations of the training/test images. Some of the orthophoto maps are also shown.



Fig. 6. Terrain surface crack images (the first and third rows) and the human expert annotated crack lines (the second and fourth rows).

B. Evaluation Metrics and Experimental Settings

1) Evaluation Metrics: There is also noise in the sinuous lines, as shown in Fig. 6. Instead, we buffered the crack lines (the buffer width of a line was set to seven pixels) and changed the lines to thin polygons, as labels. The predicted segmentation map was then compared with the ground truth after polygonization. We used the intersection over union (IoU) score to count the correctly extracted crack instances. A predicted crack polygon with an IoU score exceeding the predefined threshold (50% in our study case) was recorded as a true positive (TP), or otherwise as a false positive (FP). False negative (FN) denotes that the crack ground-truth polygon was not detected. The instance-level precision, recall, and F_1 score could then be calculated as follows:

$$\text{Precision} = \frac{\|\text{TP}\|}{\|\text{TP}\| + \|\text{FP}\|} \tag{3}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:04:21 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON OF THE DIFFERENT METHODS ON THE WHOLE MADUO CRACK DATASET

Network	F_1 *score* (%)	*Precision* (%)	*Recall* (%)	*Params* (M)	GFLOPs
UNet [35]	67.1	50.9	98.2	31.0	218.9
PSPNet [36]	64.7	48.9	95.6	49.1	184.4
SegNet [19]	66.8	51.0	96.8	29.4	160.4
RefineNet [37]	66.2	49.9	98.3	113.9	249.0
DeepLabv3+ [38]	62.2	45.7	97.6	44.5	200.0
DANet [39]	69.1	52.9	**99.6**	49.5	57.1
DeepCrack [18]	67.4	52.6	93.8	30.9	547.4
DeepCrack [21]	66.9	51.1	96.9	14.7	80.5
Crack-CADNet (Ours)	**80.7**	**69.0**	97.2	20.7	61.4

TABLE II
COMPARISON OF THE DIFFERENT METHODS ON THE MADUO CRACK DATASET LIMITED BY THE EARTHQUAKE RUPTURE REGION

Network	F_1 *score* (%)	*Precision* (%)	*Recall* (%)	*Params* (M)	GFLOPs
UNet [35]	79.6	67.2	97.8	31.0	218.9
PSPNet [36]	79.2	67.8	95.2	49.1	184.4
SegNet [19]	79.4	67.5	96.3	29.4	160.4
RefineNet [37]	83.1	72.5	97.2	113.9	249.0
DeepLabv3+ [38]	78.4	65.7	97.2	44.5	200.0
DANet [39]	80.2	67.2	**99.2**	49.5	57.1
DeepCrack [18]	80.4	70.8	93.0	30.9	547.4
DeepCrack [21]	80.0	68.4	96.2	14.7	80.5
Crack-CADNet (Ours)	**87.2**	**79.1**	97.0	20.7	61.4

$$\text{Recall} = \frac{\|\text{TP}\|}{\|\text{TP}\| + \|\text{FN}\|} \tag{4}$$

$$\text{F}_1\text{-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

where $||\cdot||$ means counting the number of all the TP (or FP, FN) instances. Precision indicates how many detected crack polygons had correspondences in the ground truth; recall indicates how many ground-truth polygons were correctly covered by the detected crack polygons; and the F_1 score provides a comprehensive metric to balance precision and recall. In practice, some adjacent cracks in images can be difficult to separate. We used morphological closing operations to merge these very close crack instances in both the prediction maps and the ground truth, with the kernel size of the closing operation equal to the buffer width of the crack lines.

2) Experimental Settings: Crack-CADNet and all the comparison methods were implemented in the PyTorch deep learning framework. Due to the limitation of GPU capacity, we cropped the DOMs into 512 × 512 image tiles with a stride of 448 pixels, i.e., a 1/8 overlapping cropping operation was employed for preparing the training dataset, in which 10% of cropped tiles are used as the validation set. In the training procedure, data augmentation strategies were used for all the methods, including random horizontal and vertical flipping, linear stretching, and histogram equalization. The initial learning rate was set to $1e^{-4}$ and divided by 10 after every 20 epochs, and the maximum number of training epochs was set to 80. The Adam optimizer [40] was employed to update the model parameters. Following the work of Liu *et al.* [21], all the methods were trained without using any pre-trained models as there are great differences in the semantic categories of previous datasets, such as MS-COCO [41] and the Maduo crack dataset. In the testing stage, the DOMs were also cropped into 512 × 512 tiles with 1/4 overlap, but in an online and engineering manner, to obtain a complete prediction map with the same size as the input DOM. The width of the crack lines in all the training and test ground-truth maps was set to seven pixels. The experiments in this study were all executed using a Windows PC equipped with a single NVIDIA GeForce GTX 1080 TI GPU, for a fair comparison.

Note that the crack detection experiments were conducted on the whole test region (natural terrain surface crack detection) and the earthquake rupture region (earthquake crack detection), and the corresponding results are listed in Tables I and II.

C. Comparison With the State-of-the-Art Methods

1) Methods: Recent methods for generic semantic segmentation and man-made object crack extraction were compared. UNet [35] was proposed for biomedical image segmentation; PSPNet [36], SegNet [19], RefineNet [37], DeepLabv3+ [38], and DANet [39] were developed for semantic segmentation of indoor and outdoor scenes in close-range natural images; and DeepCrack [18] and DeepCrack [21] were designed to detect cracks from close-range images. DeepCrack [18] fuses the convolutional features in the encoder and the decoder in a pairwise manner at the same scale, and is based on SegNet. DeepCrack [21] provides direct supervision for the features at each convolutional stage by multilevel side-output layers, and is based on an FCN [17].

2) Performance of Terrain Surface Crack Detection: Table I lists the F_1 score, precision, and recall for the eight comparison methods and the proposed Crack-CADNet method. Crack-CADNet obtains the best result of 80.7% in F_1 score and outperforms the second-best DANet by a huge margin of 11.6%. In contrast, the results of the two DeepCrack methods

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:04:21 UTC from IEEE Xplore. Restrictions apply.

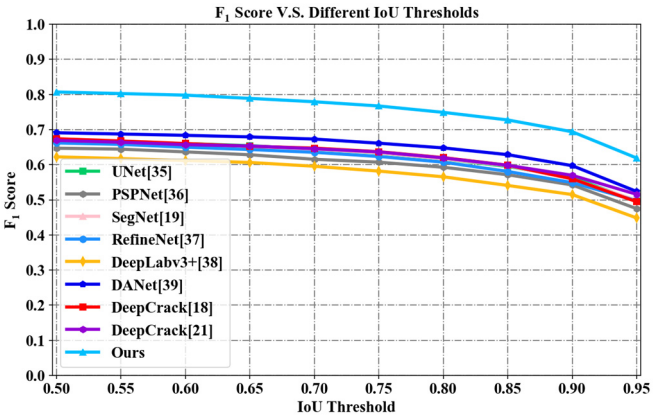


Fig. 7. F_1 curve of different crack detection methods at the varied IoU thresholds on the whole Maduo crack dataset.

(which were specially designed for crack detection), DANet, UNet, SegNet, and RefineNet show very little difference. This indicates that the proposed Crack-CADNet method, which takes the sinuous spatial shape of cracks into account, has broken through the bottleneck of the previous methods, which ignore the unique spatial property of cracks. The DeepLabv3+ method obtains the lowest F_1 score of 62.2%.

The recall rates of all the methods are very high, compared to the relatively low precision. The main reason for this is that the many confusing crack-like structures confused the algorithms. This issue is discussed in Section III-E.

We took the IoU threshold as 0.5, as used in many object detection and instance segmentation studies [42]. If more accurate predicted crack polygons are desired, the threshold could be set higher. We drew the F_1 score curves with respect to the different IoU score thresholds, which were varied from 0.5 to 0.95 at an interval of 0.05, as shown in Fig. 7. It can be observed that, at all the threshold settings, the proposed Crack-CADNet method comprehensively outperforms the other methods, demonstrating its robustness.

Examples of the crack detection results obtained by the proposed Crack-CADNet method are shown in Fig. 8. The images of the first row to the fourth row are from the test images of regions A, B, C, and D in Fig. 5, respectively. It can be observed that the soil types of the terrain surfaces are varied, and also that the cracks have complex and varied structures, widths, and shapes. It can be observed that the densities of cracks are also significantly varied in different soil types. However, the proposed Crack-CADNet method delineates all of these irregular cracks well.

It can also be observed from Table I that Crack-CADNet is very efficient. It has fewer parameters than all of the other methods, except DeepCrack [21], and its computation cost evaluated in GFLOPs is the lowest, except for DANet [39] (GFLOPs are the number in billions of floating-point operations that need to be conducted once for the model's forward propagation process).

3) Earthquake Crack Detection Performance: There is actually a fuzzy zone between earthquake cracks and nontectonic cracks in images, as shown in Fig. 9, in that the latter can be caused by drought. It is difficult for an algorithm to distinguish these cracks. Nevertheless, experts can label the actual earthquake cracks based on the knowledge of the epicenter location, the magnitude of the earthquake, and the strike of the seismic zone. Such knowledge can be used to exclude nontectonic cracks. In this study, we added a boundary for the earthquake rupture region given by experts to the test images and removed those predicted cracks that were outside the region, as shown in Fig. 10. We then compared the results of all the methods again, as shown in Table II. The precision score of each method is significantly improved as the false positive detections outside the earthquake rupture region are effectively reduced, which results in a huge improvement in F_1 score. The proposed Crack-CADNet method achieves both the best F_1 score and precision, and the precision and F_1 score in Table II are 10.1% and 6.5% higher than in Table I. All the other comparison methods perform almost the same in terms of the F_1 score, which varies from 78.4% to 80.4%, except for the heaviest RefineNet, with a higher F_1 score of 83.1%.

TABLE III

ABLATION STUDY RESULTS FOR DIFFERENT MODULE IN THE PROPOSED METHOD ON THE MADUO CRACK DATASET. DUL: DEFORMABLE UPSAMPLING LAYER, CCS: CONTEXT-CHANNEL-SPACE MODULE

Network	*F_1 score* (%)	*Precision* (%)	*Recall* (%)	*Params*(M)
DLA-34 (baseline)	67.9	52.3	96.7	18.0
DDLA-34	72.0	56.7	98.7	19.7
DDLA-34+DUL	72.7	57.5	**98.9**	20.0
DDLA-34+DUL+CCS	**80.7**	**69.0**	97.2	20.7

D. Ablation Experiments

1) Effectiveness of the Individual Parts: We verified the effectiveness of each designed module in the proposed Crack-CADNet method. We used the original DLA-34 [30] as the comparison baseline. As shown in Table III, a 4.1% F_1 score improvement is obtained by replacing DLA-34 with the deformable convolution version (i.e., DDLA-34), demonstrating that the adaptive sampling approach is much more suitable for the shape property of terrain surface cracks. The deformable upsampling layers (DULs) further enhance the performance by 0.7%, 0.8%, and 0.2% in terms of F_1 score, precision, and recall, respectively. The introduction of the context-channel-space (CCS) module with the optimal sub-module sequence obtains an 8.0% F_1 improvement. The results of other combination sequences are listed in Table IV, where all the combinations outperform the baseline by at least 2.9% in terms of F_1 score.

2) Settings of Context Aggregation Sub-Module: We tried different numbers of the parallel deformable convolution layers and different sizes/types of kernels in the context aggregation sub-module. It is observed from Tables V and VI that, four 3×3 deformable convolution groups achieved the best performance, more groups or larger kernels are not recommended.

3) Different Weights on Positive and Negative Samples: The imbalance between the foreground and background classes can hinder the training procedure. We added a weighting factor w for the foreground in the loss function (3) to strengthen

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:04:21 UTC from IEEE Xplore. Restrictions apply.

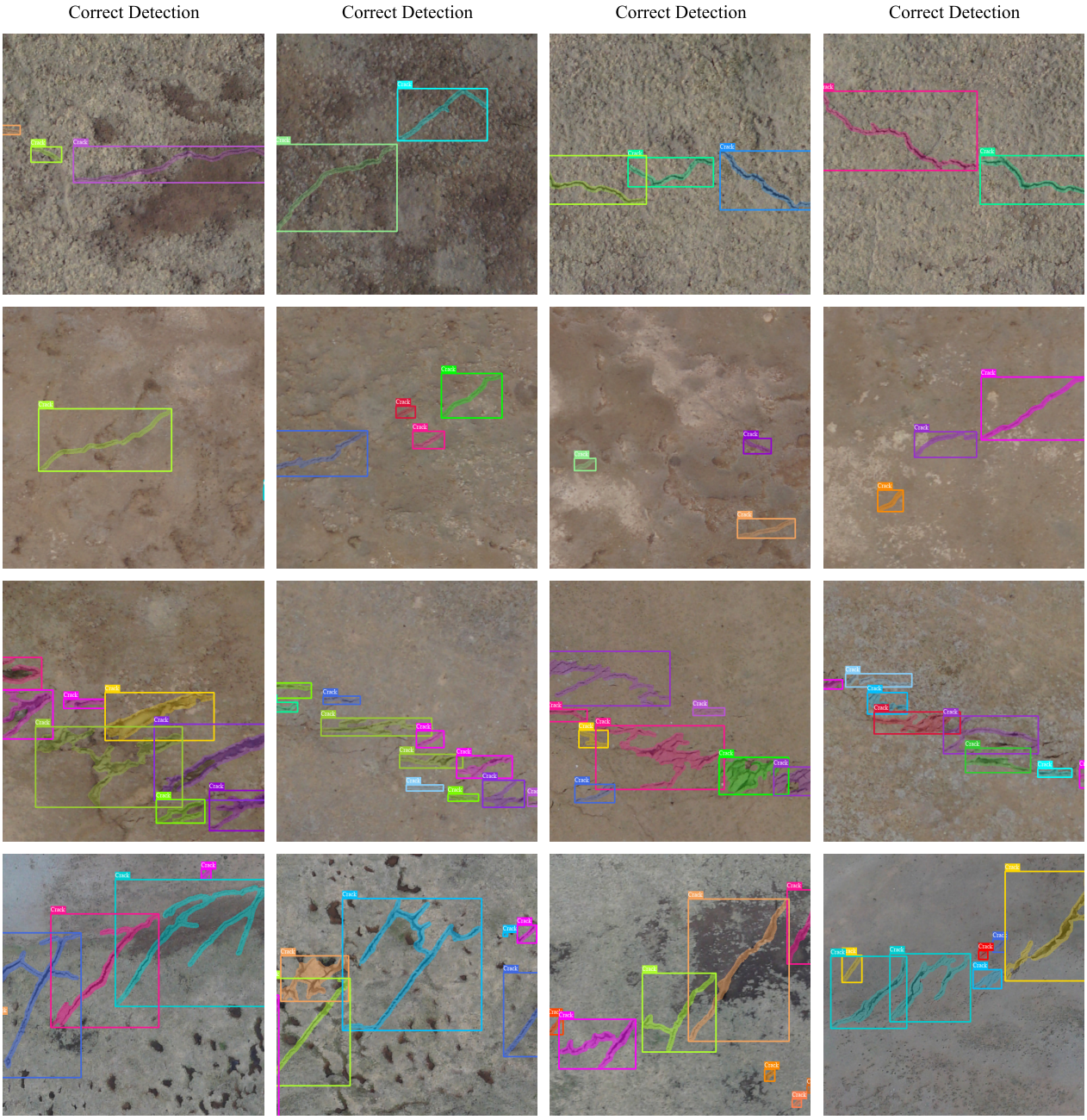


Fig. 8. Detection results of our proposed Crack-CADNet. The colorful masks covered on the images are the extracted true-positive instances. Bounding boxes are added for better readability.

the cracks. We changed the values of w to discover the best setting. As shown in Table VII, the bigger weight factors ($w = 10$ and 15) obtain higher F_1 scores. In this case, the weight factor of 10 is the best choice.

4) Buffer Width Settings: We buffered the crack lines annotated by experts to obtain thin polygons, as the buffer width can affect the performance of the algorithms. We changed the buffer width from three pixels to nine pixels at a two-pixel interval. Such a buffering operation is necessary for the deep learning-based segmentation models as we found that the models cannot converge when the linewidth remains one pixel. From Table VIII, it can be observed that the wider lines produce a higher recall but a lower precision. The linewidth of seven pixels can be observed to be the best choice in our study case, considering both the precision and recall. In addition, setting the linewidth as seven pixels also produces satisfactory visual effects, as can be seen in the examples shown in Fig. 11.

E. Discussion

1) Efficiency: The extraction efficiency of CNN methods is significantly higher than that of a manual annotation procedure. In fact, it only takes 14 s to process a 5000 × 5000-pixel image with a single NVIDIA GeForce GTX 1080 TI GPU, on average, with the proposed Crack-CADNet method. However, considering the relatively low detection precision, manual postprocessing is necessary. Even though we ignore the time

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:04:21 UTC from IEEE Xplore. Restrictions apply.



Fig. 9. Nontectonic cracks (left) and the earthquake cracks (right).

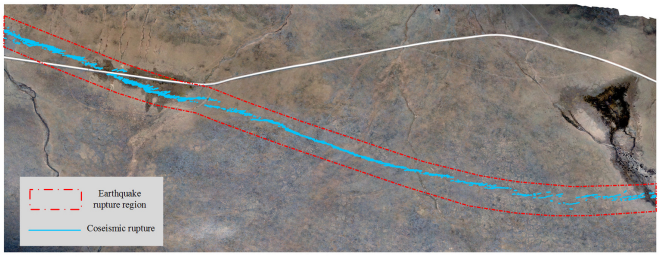


Fig. 10. Coseismic ruptures (blue lines) and the boundary of the rupture region caused by the earthquake event (red dotted line), as annotated by experts.

TABLE IV

ABLATION STUDY RESULTS OF DIFFERENT SUB-MODULE SEQUENCES IN THE FEATURE OPTIMIZATION SUB-NETWORK

Sequence	F_1 *score* (%)	*Precision* (%)	*Recall* (%)
None (baseline)	72.7	57.5	98.9
Context+Space+Channel	75.6	61.0	**99.2**
Context+Channel+Space	**80.7**	**69.0**	97.2
Channel+Context+Space	77.5	65.1	95.7
Channel+Space+Context	76.2	62.3	98.1
Space+Context+Channel	75.7	61.5	98.4
Space+Channel+Context	79.3	66.8	97.4

TABLE V

DETECTION RESULTS OF USING DIFFERENT NUMBERS OF DEFORMABLE CONVOLUTION GROUPS IN THE CONTEXT AGGREGATION SUB MODULE

Group Number	F_1 *score* (%)	*Precision* (%)	*Recall* (%)
1	75.9	62.4	96.9
2	77.5	64.2	97.6
4	**80.7**	**69.0**	97.2
6	75.5	61.2	**98.6**

TABLE VI

DETECTION RESULTS OF USING CONVOLUTION WITH DIFFERENT KERNEL SHAPES AND TYPES IN THE CONTEXT AGGREGATION SUB MODULE

Kernel Shape and Type	F_1 *score* (%)	*Precision* (%)	*Recall* (%)
[3×3+3×3]×4 (Deformable)	**80.7**	**69.0**	97.2
[5×5+5×5]×4 (Deformable)	78.5	65.0	**99.0**
[3×3+3×3]×4 (Regular)	76.0	62.6	96.6
[3×3+3×3]+ [5×5+5×5]+ [7×7+7×7]+[9×9+9×9](Deformable)	76.4	63.1	96.7

TABLE VII

DETECTION RESULTS OF THE PROPOSED METHOD USING DIFFERENT POSITIVE SAMPLE WEIGHT FACTORS DURING THE TRAINING PROCEDURE

Positive Sample Weight	F_1 *score* (%)	*Precision* (%)	*Recall* (%)
1	75.9	61.7	**98.5**
5	76.2	63.0	96.3
10	**80.7**	**69.0**	97.2
15	77.2	63.9	97.4

TABLE VIII

DETECTION RESULTS OF THE PROPOSED METHOD OBTAINED USING DIFFERENT LINEWIDTH CRACK GROUND-TRUTH MAPS DURING THE TRAINING AND EVALUATION PROCEDURES

Ground truth line width	F_1 *score* (%)	*Precision* (%)	*Recall* (%)
3 pixels	74.1	**74.3**	74.0
5 pixels	78.1	66.8	94.0
7 pixels	**80.7**	69.0	97.2
9 pixels	74.7	60.3	**98.0**

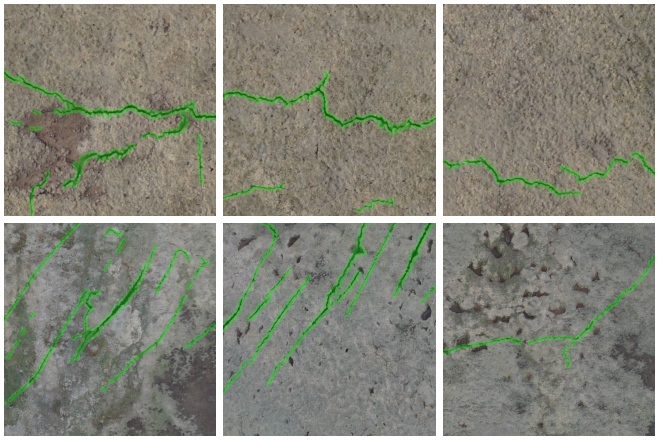


Fig. 11. Illustration of the crack annotations (transparent light green masks) generated by buffering the expert-delineated crack lines to 7-pixel-width thin polygons.

taken when browsing the whole image to find the cracks, it takes an average of 12.5 s to manually delineate a crack. In contrast, it takes only 3.2 s to delete a redundant one. Taking the test image of the experiment as an example, there are a total of 1560 ground-truth cracks and 1912 extracted crack instances obtained by Crack-CADNet in the earthquake rupture region after the buffering and morphological closing operations. There are 1513 true positive detection results (note that the IoU threshold was set to 50%, which is enough to locate these important cracks accurately in an emergency), and thus 47 cracks needed to be supplemented manually. We can therefore boost the efficiency by about nine times [i.e., $(1560 \times 12.5)/((1912 - 1513) \times 3.2 + 47 \times 12.5) - 1 \approx 9.5$]. Considering the additional time spent to search for cracks in large-size remote sensing images, it is extremely efficient to employ the combined strategy of employing an

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:04:21 UTC from IEEE Xplore. Restrictions apply.



Fig. 12. Detection results of the proposed Crack-CADNet method. The first and third columns are the original images, and the red polygons in the second and the fourth columns are the false positive instances.

automatic CNN extraction method and subsequent manual postprocessing, especially when the crack map is required urgently after a sudden earthquake event.

2) Limitations: Although the proposed method outperformed all of the comparison methods, the proposed method only obtained a precision of 69.0% when detecting all the terrain surface cracks and 79.1% when excluding the extraction results located in the nonearthquake rupture regions. In contrast, the recall was 97% in both cases. After a close check, we found there are many crack-like structures that impact the algorithm. As shown in Fig. 12, the topographic changes form edges (the first row) and dry rivers also make boundaries (the second row), which are both difficult to separate from earthquake cracks. In addition to introducing the knowledge of experts, postprocessing is also required to eliminate the falsely detected results. Except for manual checking, other automated methods could be tried. For example, a comparison of pre- and post-earthquake high-resolution digital surface models (DSMs) could be used to exclude some topography-related errors, or land-use/land-cover maps could be used to exclude cracks produced by human activities.

3) Potential Applications: An earthquake crack map obtained after an earthquake can be used by a seismologist to study the subsequent fault activity, and to predict whether further earthquakes or landslides are likely to be triggered. At the same time, it is very important for local government to evacuate residents from the high-risk areas and to identify damaged roads and bridges in good time. The crack maps that can be rapidly produced using the proposed method could help in this regard. For example, the residents in the high-risk areas could be quickly located by overlaying the crack map with the GIS map of residential buildings. More potential applications in reducing disaster losses and investigating earthquakes and tectonic movements will be investigated in our future work.

IV. Conclusion

In this article, we have proposed an automatic deep learning-based pipeline for earthquake crack extraction, which was tested using the Maduo earthquake as a case study. The preprocessing includes data acquisition with low-altitude UAVs, DOM generation with photogrammetric technology, and manual crack map delineation. The core is a novel context-channel-space boosted deformable convolutional network (Crack-CADNet), which considers the shape property of terrain surface cracks and outperforms the recent generic and crack detection specific semantic segmentation methods that ignore this property. Postprocessing according to human expert knowledge is also incorporated to exclude the nontectonic cracks outside of the rupture region and obtain an instance-level earthquake crack map.

We also established that the variety of complex topographic structures is the main factor that affects the precision of earthquake crack extraction. Nevertheless, this study is the first comprehensive work to have extracted earthquake cracks using deep learning. There is room for further improving both the pipeline and the network. We hope that this work will serve as effective technical support for geologists to study the causes of earthquakes, monitor active fault movement, and provide early warning of latent earthquakes or other disasters.

References

[1] W. Wang *et al.*, "Aftershock sequence relocation of the 2021 MS7.4 Maduo earthquake, Qinghai, China," *Sci. China Earth Sci.*, vol. 64, no. 8, pp. 1371–1380, Aug. 2021.

[2] K. Zhang, H. D. Cheng, and B. Zhang, "Unified approach to pavement crack and sealed crack detection using preclassification based on transfer learning," *J. Comput. Civil Eng.*, vol. 32, no. 2, Mar. 2018, Art. no. 04018001.

[3] F. Yang, L. Zhang, S. Yu, D. V. Prokhorov, X. Mei, and H. Ling, "Feature pyramid and hierarchical boosting network for pavement crack detection," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 4, pp. 1525–1535, Apr. 2019.

[4] S. Dorafshan, R. J. Thomas, and M. Maguire, "Fatigue crack detection using unmanned aerial systems in fracture critical inspection of steel bridges," *J. Bridge Eng.*, vol. 23, no. 10, Oct. 2018, Art. no. 04018078.

[5] X. Wu and X. Liu, "Building crack identification and total quality management method based on deep learning," *Pattern Recognit. Lett.*, vol. 145, pp. 225–231, May 2021.

[6] S. Krishnamoorthy, C. Zhang, and Z. Yanxin, "Implementation of image fusion to investigate wall crack," in *Proc. Int. Conf. Emerg. Trends Inf. Technol. Eng. (ic-ETITE)*, Feb. 2020, pp. 1–3.

[7] C. Feng, M.-Y. Liu, C.-C. Kao, and T.-Y. Lee, "Deep active learning for civil infrastructure defect detection and classification," in *Proc. ASCE Int. Workshop Comput. Civil Eng.*, Jun. 2017, pp. 298–306.

[8] C. Koch, K. Georgieva, V. Kasireddy, B. Akinci, and P. Fieguth, "A review on computer vision based defect detection and condition assessment of concrete and asphalt civil infrastructure," *Adv. Eng. Inform.*, vol. 29, no. 2, pp. 196–210, Apr. 2015.

[9] Y. Yuan *et al.*, "Crack length measurement using convolutional neural networks and image processing," *Sensors*, vol. 21, no. 17, p. 5894, Sep. 2021.

[10] T. Nishikawa, J. Yoshida, T. Sugiyama, and Y. Fujino, "Concrete crack detection by multiple sequential image filtering," *Comput.-Aided Civil Infrastruct. Eng.*, vol. 27, no. 1, pp. 29–47, 2012.

[11] P. Prasanna *et al.*, "Automated crack detection on concrete bridges," *IEEE Trans. Autom. Sci. Eng.*, vol. 13, no. 2, pp. 591–599, Apr. 2014.

[12] W. Zhang, Z. Zhang, D. Qi, and Y. Liu, "Automatic crack detection and classification method for subway tunnel safety monitoring," *Sensors*, vol. 14, no. 10, pp. 19307–19328, 2014.

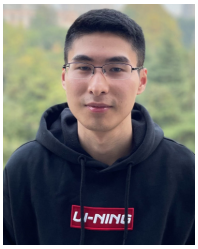
[13] C. Han, T. Ma, J. Huyan, X. Huang, and Y. Zhang, "CrackW-Net: A novel pavement crack image segmentation convolutional neural network," *IEEE Trans. Intell. Transp. Syst.*, early access, Aug. 30, 2021, doi: 10.1109/TITS.2021.3095507.

[14] X. Yang, H. Li, Y. Yu, X. Luo, T. Huang, and X. Yang, "Automatic pixel-level crack detection and measurement using fully convolutional network," *Comput.-Aided Civil Infrastruct. Eng.*, vol. 33, no. 12, pp. 1090–1109, Dec. 2018.

[15] L. F. Zhang Yang *et al.*, "Road crack detection 12 > replace this line with your manuscript ID number (double-click here to edit) <using deep convolutional neural network," in *Proc. Int. Conf. Image Process.*, 2016, pp. 3708–3712.

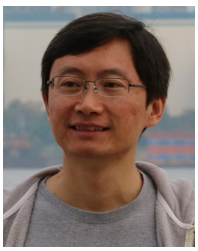
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:04:21 UTC from IEEE Xplore. Restrictions apply.

[16] L. Pauly, H. Peel, S. Luo, D. Hogg, and R. Fuentes, "Deeper networks for pavement crack detection," in *Proc. Int. Symp. Autom. Robot. Construct. (IAARC)*, Jul. 2017, pp. 479–485.
[17] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.
[18] Q. Zou, Z. Zhang, Q. Li, X. Qi, Q. Wang, and S. Wang, "DeepCrack: Learning hierarchical convolutional features for crack detection," *IEEE Trans. Image Process.*, vol. 28, no. 3, pp. 1498–1512, Mar. 2018.
[19] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder–decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Jan. 2017.
[20] C. V. Dung and L. D. Anh, "Autonomous concrete crack detection using deep fully convolutional neural network," *Autom. Construct.*, vol. 99, pp. 52–58, Mar. 2019.
[21] Y. Liu, J. Yao, X. Lu, R. Xie, and L. Li, "DeepCrack: A deep hierarchical feature learning architecture for crack segmentation," *Neurocomputing*, vol. 338, pp. 139–153, Apr. 2019.
[22] H. Li *et al.*, "SCCDNet: A pixel-level crack segmentation network," *Appl. Sci.*, vol. 11, no. 11, p. 5074, May 2021.
[23] J. Huyan, W. Li, S. Tighe, Z. Xu, and J. Zhai, "CrackU-Net: A novel deep convolutional neural network for pixelwise pavement crack detection," *Struct. Control Health Monitor.*, vol. 27, no. 8, p. e2551, Aug. 2020.
[24] Z. Liu, Y. Cao, Y. Wang, and W. Wang, "Computer vision-based concrete crack detection using U-Net fully convolutional networks," *Autom. Construct.*, vol. 104, pp. 129–139, Dec. 2019.
[25] F. Ni, J. Zhang, and Z. Chen, "Pixel-level crack delineation in images with convolutional feature fusion," *Struct. Control Health Monitor.*, vol. 26, no. 1, p. e2286, Jan. 2019.
[26] A. Zhang *et al.*, "Automated pixel-level pavement crack detection on 3D asphalt surfaces using a deep-learning network," *J. Comput.-Aided Civil Infrastruct. Eng.*, vol. 32, no. 10, pp. 805–819, 2017.
[27] Y. Bai, B. Zha, H. Sezen, and A. Yilmaz, "Engineering deep learning methods on automatic detection of damage in infrastructure due to extreme events," *Struct. Health Monitor.*, early access, May 2022, Art. no. 147592172210836, doi: 10.1177/14759217221083649.
[28] J. Dai *et al.*, "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 764–773.
[29] X. Ke, X. Zhang, and T. Zhang, "GCBANet: A global context boundary-aware network for SAR ship instance segmentation," *Remote Sens.*, vol. 14, no. 9, p. 2165, Apr. 2022.
[30] F. Yu, D. Wang, E. Shelhamer, and T. Darrell, "Deep layer aggregation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2403–2412.
[31] X. Zhou, D. Wang, and P. Krähenbühl, "Objects as points," 2019, *arXiv:1904.07850*.
[32] S. Woo *et al.*, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19.
[33] J. Hu, L. Shen, and G. Sun, "Squeeze- and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.
[34] S. Ji, D. Yu, C. Shen, W. Li, and Q. Xu, "Landslide detection from an open satellite imagery and digital elevation model dataset using attention boosted convolutional neural networks," *Landslides*, vol. 17, no. 6, pp. 1337–1352, Jun. 2020.
[35] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241.
[36] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.
[37] G. Lin, A. Milan, C. Shen, and I. Reid, "RefineNet: Multi-path refinement networks for high-resolution semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1925–1934.
[38] L. C. Chen *et al.*, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 801–818.
[39] J. Fu *et al.*, "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.
[40] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
[41] X. Chen *et al.*, "Microsoft COCO captions: Data collection and evaluation server," 2015, *arXiv:1504.00325*.
[42] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8759–8768.



Dawen Yu received the B.S. degree from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2018, where he is currently pursuing the Ph.D. degree.

His research interests include remote sensing image processing and machine learning.



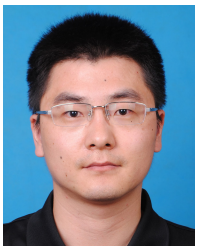
Shunping Ji (Senior Member, IEEE) received the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2007.

He is currently a Professor with the School of Remote Sensing and Information Engineering, Wuhan University. His research interests include photogrammetry, remote sensing image processing, mobile mapping systems, and machine learning.



Xue Li received the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, Hubei, China, in 2010.

He is currently an Associate Researcher with the Institute Seismology, China Earthquake Administration, Wuhan. His research interests include remote sensing disaster mitigation applications and remote sensing monitoring methods for social resilience of earthquake disasters.



Zhaode Yuan received the Ph.D. degree in active tectonic and paleoseismology from the Institute of Geology, China Earthquake Administration, Beijing, China, in 2018.

He is currently an Associate Researcher with the Institute of Geology, China Earthquake Administration. His research interests include coseismic surface rupture, paleoseismology, and geologic slip rates along large-scale strike-slip fault.



Chaoyong Shen received the Ph.D. degree in forestry equipment and informatization from Beijing Forestry University, Beijing, China, in 2020.

He is currently a Research Professor with The Third Surveying and Mapping Institute of Guizhou Province, Guiyang, China. His research interests include remote sensing image processing and UAV photogrammetry.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:04:21 UTC from IEEE Xplore. Restrictions apply.